

DHEERAJ KARKI

Software Engineer | C++ · CAD · Robotics

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PROFESSIONAL SUMMARY

Software engineer with 10+ years building C++ applications in the CAD industry, specialising in computational geometry, spatial data structures, system performance, and cross-platform interactive applications. Currently pursuing an MSc in Robotics at the University of Bristol to apply that foundation to autonomous and intelligent systems.

TECHNICAL SKILLS

Languages: C++, C, JavaScript, Python

Domains: Computational geometry, spatial data structures, CAD systems, geometry processing, rendering & graphics, algorithms

Frameworks & Libraries: React JS, ODA (Open Design Alliance)

Concepts: Planar graph traversal, closed-boundary detection, cross-platform development, system-level optimisation, real-time interactive applications

Practices: Software architecture, code refactoring, performance optimisation, developer mentoring

PROFESSIONAL EXPERIENCE

Arden Software Hyderabad, India

Jan 2019 – Present

Staff Software Developer

Sep 2025 – Present

- Contributed to migration of a traditional desktop CAD application into a modern engine running across web, desktop, and cloud, delivering feature parity from the legacy system to the new platform.
- Designed spatial data structures (quad-trees) for polygon lookup, point-in-polygon location, and nested polygon-hierarchy resolution, accelerating geometric queries by ~35%.
- Contribute to software architecture for the CAD team and support the UX team on interface design, alongside code refactoring and mentoring 4–5 developers on computational geometry and algorithms.

Senior Software Developer

Oct 2021 – Sep 2025

- Developed the annotation framework for the CAD drawing engine.
- Designed algorithms for planar graph traversal and closed-boundary detection in complex drawings.
- Developed import/export support across four CAD formats — DXF, DWG, IPD, and an internal JSON format — improving cross-tool interoperability.
- Delivered rendering optimisations and geometry-processing tools that improved platform performance and usability.

Software Developer

Jan 2019 – Oct 2021

- Contributed to a cloud-based CAD system supporting interactive design workflows.
- Improved canvas rendering performance by ~90% on heavy CAD drawings by intercepting render calls, replaying cached geometry to avoid recomputation, and batching draw operations.

Graebert GmbH Noida, India

Feb 2016 – Jan 2019

Senior Software Engineer

Jan 2018 – Jan 2019

- Delivered multiple features for the Onshape Drawings team on a cloud-based CAD platform, including dimensioning, shaded views, and the drawing-property panel.
- Built extensible plugin frameworks and drawing-property systems enabling modular CAD features.
- Collaborated within a 20-person distributed team across Germany, Russia, and India.

Software Engineer

Feb 2016 – Jan 2018

- Contributed to a cross-platform cloud CAD editor using C++, JavaScript, Qt, and ODA libraries.

- Developed custom entities, view systems, and interactive editing tools; improved rendering and interaction performance.

Sevya Multimedia Technology Hyderabad, India

Feb 2014 – Jan 2016

Software Developer

- Consulted for clients including STMicroelectronics and Aristocrat, plus internal projects, building a C/C++ automation tool that replaced a manual workflow and contributing to an embedded home-automation prototype for customer demonstrations.

Electronics Corporation of India Limited (ECIL) Dehradun, India

Mar 2013 – Jan 2014

Engineer Consultant

- Led a team of 10 operators on the National Population Register project, processing ~200 records per day and applying data-protection and encryption practices for sensitive citizen data.

PROJECTS & VENTURES

Frido · frido.app Side venture

Jul 2019 – Jul 2022

Co-owner & Developer

- Led end-to-end development of a productivity application designed to reduce information overload and help users focus on one task at a time.
- Collaborated closely with designers to translate UX concepts into functional, minimalist interfaces focused on usability.
- Launched on the Apple ecosystem to a 5-star rating and reached 2nd Product of the Day on Product Hunt.

EDUCATION

MSc Robotics — University of Bristol, UK

2026 – 2027 (expected)

M.Tech in Software Systems — BITS Pilani, India

Jan 2017 – Jan 2019

Specialisation: Embedded Systems. CGPA 8.6/10. Dissertation: "CAD in Cloud."

B.Tech in Electronics & Communication — ICFAI University, Dehradun, India

Aug 2008 – May 2012

CGPA 9.42/10. Thesis: "Accelerated Parallel Processing Using OpenCL." Study project: Wireless Power Transmission.

ACHIEVEMENTS & AWARDS

- Silver Medal (2nd Rank) — B.Tech, ICFAI University (2012).
- Merit Fellowship — 50% tuition waiver throughout B.Tech, ICFAI University (2008–2012).
- First Division — M.Tech, BITS Pilani (2018).